A PRACTITIONER'S FIELD GUIDE · BEGINNER TO INDUSTRY EXPERT

Hedge Fund Treasury, Funding & Financing

How a fund finances its book — prime brokerage, margin, repo, swaps and collateral — and why the financing desk is a genuine, repeatable source of alpha. From first principles to the frontier.

Treasury & Financing Masterclass
Edition 1.0 · 2026.10
Companion to the Treasury cinematic course

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01 · EXECUTIVE SUMMARY

How to Use This Guide & Executive Summary

At a hedge fund, the portfolio manager chooses what to own. The financing desk decides whether owning it is worth it — because every long must be funded, every short must be borrowed, and every derivative must be margined. On a levered book, shaving a handful of basis points off financing or freeing a slice of trapped margin is pure, repeatable alpha that no market call is needed to earn. This guide takes you from "what is a prime broker" to running a multi-prime financing book like a managed P&L.

Who this is for

Anyone who wants to understand — and speak fluently about — how a hedge fund actually finances itself: analysts and treasurers joining a fund; prime-brokerage, delta-one and securities-financing professionals on the sell side; allocators and risk officers assessing a manager's funding resilience; and technologists building the systems that run margin, collateral and cash. No prior knowledge is assumed; every term is defined on first use.

How the guide is built

The twelve chapters climb a ladder. Chapters 2–3 are the foundations: why funding is alpha, and the prime-brokerage relationship that sits at the centre of everything. Chapters 4–6 are the three ways a fund gets leverage: margin (Reg-T and portfolio margin), repo and securities financing, and synthetic financing via swaps and total-return swaps. Chapters 7–9 are the advanced disciplines: derivatives margin (cleared and uncleared), multi-prime and treasury operations, and liquidity and margin-stress risk. Chapters 10–12 cover crisis history and mastery — detailed post-mortems (Archegos, LTCM, March 2020, the 2022 LDI spiral), then a full field catalogue of a century of liquidity crunches with their calculations, and finally what a best-in-class financing desk looks like. Chapter 13 is a glossary and reading list.

HOW TO READ IT

Read linearly the first time — each chapter builds on the last. Every cost or margin idea is shown as a fully worked calculation with real numbers, every new term sits in a definition box, and every chapter ends with a "Best in class" box describing what elite financing desks actually do. Afterwards it works as a reference you can pull a single number or method from.

The one-paragraph version

A hedge fund holds more exposure than it has capital, so it must borrow cash to carry longs, borrow stock to sell shorts, and post margin to trade derivatives. It gets that leverage three ways: from its prime broker as a margin loan, in the repo market against bonds, or synthetically through swaps. Each has a price (a spread over SOFR, a haircut, a borrow fee) and a margin cost (Reg-T, portfolio margin, SIMM, CCP initial margin). The

financing desk's job is to source that leverage at the lowest all-in cost, spread it across multiple prime brokers to manage counterparty risk, hold enough unencumbered cash to survive a margin spike, and never be the fund whose lenders pull the plug. Do it well and financing adds quiet, compounding alpha; do it badly and you become the next war story in Chapter 10.

What you will be able to do by the end

- Build a fund's financing P&L — margin interest, short rebate, borrow fees and cash interest.
- Compare the all-in cost of financing a position via margin loan, repo, or a total-return swap, and say which wins.
- Compute margin under Reg-T versus portfolio margin and quantify the capital it frees.
- Work a repo haircut and carry, a SPAN or SIMM margin number, and a counterparty-exposure netting.
- Run a margin-stress test and a liquidity-coverage check against redemptions and calls.
- Explain, in financing terms, exactly what broke at Archegos, LTCM, in March 2020 and in the 2022 LDI crisis — and what best practice prevents it.

QUESTIONS THIS GUIDE ANSWERS

How does a hedge fund actually get leverage, and what does each source cost? Why do sophisticated funds use swaps instead of simply borrowing? What is the difference between Reg-T and portfolio margin, and why does it matter so much to returns? How can a fund with good positions still be destroyed by a margin call? And what separates a financing desk that merely keeps the lights on from one that is genuinely world-class?

02 · FOUNDATIONS

Why Funding Is Alpha

A hedge fund does not earn its living on cash. It earns it on borrowed money and borrowed securities, applied to a book many times larger than its own capital. On that leveraged frame, the cost of financing is not an overhead to be tolerated — it is a line of return the treasury desk can widen or shrink, basis point by basis point, every single day.

What a hedge-fund treasury desk actually does

A hedge-fund treasury — often called the financing desk, funding desk, or securities-finance team — is the plumbing beneath every trade the portfolio managers put on. The portfolio manager decides what to own and short. The treasury desk decides how it is paid for, who holds it, and at what rate. Four responsibilities define the mandate.

- Finance the longs. Every long position the fund cannot pay for in full with its own cash must be financed — borrowed against on margin, repo'd, or carried synthetically through a swap. The desk sources that cash at the lowest achievable spread.
- Source the shorts. To sell a security short, the fund must first borrow it. The desk secures the locate, negotiates the borrow fee, and monitors recall and buy-in risk on every name.
- Manage cash, margin, and collateral. The desk keeps initial and variation margin posted at every prime broker, futures clearer, and swap counterparty, and it meets each call with the cheapest eligible asset.
- Control counterparty exposure. The desk watches the health of each prime broker and dealer, spreads assets across multiple relationships, and negotiates segregation and rehypothecation limits so a single failure cannot trap the book.

> Financing alpha — the portion of a fund's return that comes not from picking the right assets but from financing those assets more cheaply than a rival running the identical book. It is structural, repeatable, and uncorrelated with the investment thesis.

How a fund treasury differs from a bank treasury

Both desks are called "treasury," but they solve opposite problems. A bank treasury funds a balance sheet it is legally required to carry and is governed by liquidity ratios — the LCR and NSFR — that force it to hold stable funding and high-quality liquid assets. Its job is defensive: survive a run, satisfy the regulator, price funding internally through Funds Transfer Pricing. A hedge-fund treasury carries almost no regulated balance sheet of its own; it rents balance sheet from prime brokers and dealers. Its job is offensive: squeeze the financing spread, maximise exposure per dollar of posted margin, and turn the funding stack into return. The bank treasury worries about not failing. The fund treasury worries about not leaving money on the table.

Dimension	Bank treasury	Hedge-fund treasury
Balance sheet	Owns it; must fund it	Rents it from PBs/dealers
Primary goal	Survive stress; satisfy LCR/NSFR	Minimise financing spread; maximise leverage
Key constraint	Regulatory liquidity ratios	Margin rules + counterparty terms
Funding sources	Deposits, CP, CDs, repo	Margin loans, repo, TRS/swaps, stock borrow
Internal pricing	FTP to each desk	Financing P&L attributed to each book

The financing P&L

Everything the treasury touches resolves into a single statement: the financing P&L. For a long/short equity book it has four moving parts.

- Margin interest paid on the financed portion of the longs — a cost, quoted as a spread over SOFR.
- Short rebate received — interest the stock lender pays on the cash the fund posts against its borrows. Positive for general-collateral names, negative (a fee) for specials.
- Stock-borrow fees paid on hard-to-borrow names — a cost, quoted in annualised basis points or percent.
- Cash interest earned on the fund's free credit balances and excess margin.

> Basis point (bp) — one hundredth of one percent, 0.01%. Financing spreads live almost entirely in basis points; 25bps on a large levered book is a meaningful, bankable number.

> Gross vs net exposure — gross is long market value plus the absolute value of short market value, measured against capital (the gross leverage ratio). Net is long minus short, the directional market bet. A market-neutral fund can run 4× gross while holding net near zero.

Why shaving 25–50bps is real alpha

The leverage multiplier is the whole point. Financing cost is charged on *borrowed* dollars, but the saving flows back to the fund's own *capital*. The more leveraged the book, the larger the magnification of any spread improvement onto the return on equity.

WORKED EXAMPLE

A market-neutral fund runs $1,000mm of capital at 4× gross — $2,000mm long and $2,000mm short, $4,000mm of gross exposure. Assume the financed long book is $2,000mm carried on a margin loan.

Step 1 — annual financing cost at the old spread. SOFR = 5.00%; the desk pays SOFR + 60bps = 5.60% on the $2,000mm of longs.
$2,000mm × 5.60% = $112.0mm/yr.

Step 2 — negotiate the spread down 50bps to SOFR + 10bps = 5.10%.
$2,000mm × 5.10% = $102.0mm/yr.

Step 3 — the saving. $112.0mm − $102.0mm = $10.0mm/yr. (Equivalently, 0.50% × $2,000mm = $10.0mm.)

Step 4 — translate to return on capital. $10.0mm ÷ $1,000mm of capital = +100bps (1.00%) added to the fund's net return, every year, with zero additional market risk.

A 50bp financing win on a 2× financed-to-capital ratio becomes a full point of return on equity. That is why the desk fights for basis points.

A simple financing-P&L statement

WORKED EXAMPLE

The same fund: $2,000mm longs financed, $2,000mm shorts. SOFR = 5.00%.

Line	Basis	Rate	Annual $
Margin interest paid (longs)	$2,000mm	SOFR + 60bps = 5.60%	(112.0mm)
Short rebate received (GC shorts)	$1,600mm	SOFR − 20bps = 4.80%	+76.8mm
Borrow fees paid (specials)	$400mm	350bps fee	(14.0mm)
Cash interest earned (free cash)	$300mm	SOFR − 10bps = 4.90%	+14.7mm
Net financing P&L			(34.5mm)

The book costs $34.5mm/yr to finance before any investment return — roughly 345bps on the $1,000mm of capital, or 86bps on the $4,000mm gross. Every line is a lever: tighten the margin spread, convert specials to cheaper borrow, or sweep idle cash into SOFR-earning balances, and the drag shrinks directly into return.

Cash versus synthetic financing — the high-level choice

The desk can carry a position two ways. In cash financing the fund owns the asset and borrows against it (a margin loan, or repo for bonds); the position sits on the fund's balance sheet. In synthetic financing a dealer owns the asset and the fund holds a contract — a total return swap or CFD — that passes it the economic return in exchange for a financing leg. Synthetic is balance-sheet-light: the asset never touches the fund's books, which eases leverage limits, simplifies operations, and can open markets the fund cannot access directly. Chapters 03–06 price both routes line by line.

The counterparties

- Prime broker (PB). The hub bank that bundles custody, clearing, margin financing, securities lending, and reporting for equities and most cash positions.
- Futures commission merchant (FCM). The clearing member that carries the fund's listed futures and options and posts SPAN margin to the exchange clearing house.
- Custodian. The institution that holds the fund's assets in safekeeping, often legally separate from the financing PB for segregation.
- Repo dealer / sponsoring member. The counterparty — or FICC sponsor — that lends cash against the fund's bonds in the repo market.

BEST IN CLASS

The best financing desks treat the funding stack as a portfolio to be optimised, not a bill to be paid. They run multi-prime (three or more live, funded relationships) so no single PB can halt the book or dictate terms; they attribute a daily financing P&L to every strategy book so a PM sees the true all-in cost of a crowded short; and they benchmark every spread against what a rival of their size and balance-sheet value could command — then renegotiate annually. They treat 25bps the way a trader treats a mispriced option: a real, recurring edge worth the effort to capture.

03 · PRIME BROKERAGE

The Core Relationship

The prime broker is the single most important counterparty most hedge funds have. It custodies the assets, clears the trades, lends the cash that finances the longs, and sources the stock that covers the shorts — all under one bundled, cross-collateralised relationship. Understanding how a PB prices that relationship, and how to negotiate it, is the first discipline of a financing desk.

What a prime broker provides

> Prime broker (PB) — the bank that services a hedge fund through one integrated account, standing between the fund and the wider market for custody, financing, securities lending, clearing, capital introduction, and consolidated reporting.

- Custody and asset servicing. Holds the fund's securities and cash, processes corporate actions, dividends, and coupons.
- Clearing and settlement. Settles trades the fund executes away with other brokers (give-up or step-out), netting them into one account.
- Margin financing. Lends cash against the longs so the fund can run leverage.
- Securities lending. Locates and borrows shares so the fund can sell short.
- Capital introduction ("cap intro"). Introduces the fund to potential allocators — a soft service PBs use to win and keep mandates.
- Consolidated reporting. Delivers one daily view of positions, cash, margin, financing, and stock-loan activity.

The financing spread

A PB does not quote an absolute rate; it quotes a spread to SOFR. On the long side the fund *pays* SOFR plus a spread on the margin loan. On the short side the fund posts cash and *receives* a rebate — the short-side analogue, quoted as SOFR minus a spread for easy names, or as an outright fee for scarce ones.

> SOFR — the Secured Overnight Financing Rate, the US near-risk-free benchmark built from actual overnight Treasury repo trades. Replaced USD LIBOR as the reference rate for financing.

> Financing spread — the markup (longs) or markdown (shorts) a PB charges over/under SOFR. It compensates the PB for its balance-sheet use, funding cost, operational cost, and target return on the relationship.

Long financing — the margin loan

> Margin loan — the cash a PB lends against a long position. The fund posts the asset as collateral and pays interest at SOFR + spread on the borrowed amount. The financed amount equals the position value less the fund's own equity (the margin).

WORKED EXAMPLE

A fund buys $50mm of a large-cap stock. Under portfolio margin it posts $10mm of equity; the PB lends the remaining $40mm at SOFR + 40bps, with SOFR = 5.00% → 5.40%. Hold one year.

Step 1 — margin-loan interest. $40mm × 5.40% = $2.160mm/yr.

Step 2 — daily accrual (ACT/360). $40mm × 5.40% × (1/360) = $6,000/day.

The fund pays $2.160mm to carry $50mm of exposure on $10mm of its own cash. Cut the spread to SOFR + 15bps and the cost falls to $40mm × 5.15% = $2.060mm — a $100k/yr saving on one position.

Short financing — rebate, GC, and specials

To short, the fund borrows the stock and posts cash collateral (typically 102% of market value) with the lender. The lender invests that cash and pays most of the earnings back as the rebate.

> Rebate rate — the interest a stock lender pays the short-seller on the cash collateral posted against a borrow. A positive rebate is income to the fund; a negative rebate means the fund pays to keep the borrow.

> General collateral (GC) — stocks that are easy and plentiful to borrow. They lend near the market rate, so the rebate sits just below SOFR (e.g. SOFR − 20bps).

> Special (hard-to-borrow) — a stock in short supply relative to borrow demand. Lenders charge a fee; the rebate turns negative, sometimes deeply so for crowded shorts.

WORKED EXAMPLE

(a) GC short — positive rebate. The fund shorts $30mm of a GC name. SOFR = 5.00%; the rebate is SOFR − 25bps = 4.75% on the posted cash.
$30mm × 4.75% = +$1.425mm/yr of rebate income.

(b) Special — negative rebate. The fund shorts $30mm of a hard-to-borrow name quoted at a 400bps borrow fee. Mechanically the rebate is SOFR − 400bps = 5.00% − 4.00% = +1.00%, so the fund still earns a little on the cash:
$30mm × 1.00% = +$0.30mm/yr. If the fee had been 600bps, the rebate would be −1.00% and the fund would *pay* $0.30mm.

WORKED EXAMPLE

All-in cost of shorting a hard-to-borrow name. The fund shorts $20mm of a stock at a 400bps borrow fee. The fee is simply the drag versus holding the cash at SOFR.

Step 1 — borrow fee. $20mm × 4.00% = $0.80mm/yr paid to the lender.

Step 2 — the hurdle. The short carries a 4.00% annual cost of borrow. The thesis must deliver more than 4.00% of downside per year *just to break even on financing*, before any alpha. A 10% expected decline over 12 months becomes a 6% net short return after borrow.

Borrow cost is not a footnote — at 400bps it is a hard hurdle the PM must clear, and at the 25–50% fees seen on genuine squeeze names it can invalidate the trade entirely.

The PB's balance-sheet constraint and post-Archegos pricing

A PB does not conjure financing from nothing. Every margin loan it extends and every borrow it internalises consumes its own balance sheet — leverage-ratio capacity, funding, and regulatory capital under Basel rules. The PB prices each client on the return it earns per unit of balance sheet consumed: a client that posts lots of cash, lends out hard-to-borrow inventory the PB can re-lend, and trades actively is cheap to finance; a client that only draws margin loans and holds low-turnover positions is expensive.

> Archegos (2021) — a family office that built enormous concentrated single-name exposure synthetically across several prime brokers, each blind to the others. Its collapse cost PBs more than $10bn. The aftermath tightened PB risk standards industry-wide: stricter concentration add-ons, deeper position transparency demands, higher initial margin on single-name swaps, and far more selective client onboarding.

Post-Archegos, PBs price balance sheet harder and scrutinise concentration and leakage across counterparties. A fund that was waved through in 2019 now faces larger add-ons for the same book — making the negotiation below more valuable, not less.

Setting up and negotiating a PB agreement

The economics live in the terms. The levers a desk negotiates:

- The financing spread over SOFR on longs and the rebate schedule on shorts — the headline number.
- Margin methodology (Reg T vs portfolio margin) and the size of house add-ons for concentration and liquidity.
- Rehypothecation cap and segregation — how much of the fund's collateral the PB may re-use, and how much sits in protected accounts.
- Term and notice. Committed-term financing (e.g. 90–180 day locks) versus overnight that the PB can reprice or pull.
- Stock-loan economics — whether the fund shares in the lending revenue on its long inventory the PB re-lends.

BEST IN CLASS

The best-run funds negotiate from data and competition. They run multiple primes so each knows it can lose the book, and they bring a quantified picture of the balance-sheet value they deliver — cash balances, re-lendable long inventory, trading turnover — to argue the PB earns its return without a fat spread. They lock term financing on their core positions so a PB cannot reprice them in a stress, cap re-hypothecation tightly, and keep excess margin segregated. The goal: resilient, transparent, cheap financing that survives the next Archegos-style tightening rather than being its victim.

04 · MARGIN METHODOLOGIES

How Margin Is Measured

Margin is the collateral a fund posts to support a leveraged position, and the method used to size it is one of the biggest determinants of how much capital a strategy consumes. The same hedged book can require vastly different margin under a flat percentage rule versus a risk-based model. This chapter walks the main methodologies, the capital they free, and the risks that freedom carries.

Regulation T — the flat percentage

> Regulation T (Reg T) — the US Federal Reserve rule setting a flat initial margin of 50% on most marginable equities (so at most 2× leverage on a long) and a maintenance margin of 25%. Reg T treats each position in isolation — it grants no credit for one position hedging another.

Reg T is simple and punitive for hedged books. A long/short pair that is economically near-flat still attracts 50% margin on the long and a separate requirement on the short, because the rule never nets the two. For a classic long/short fund, Reg T strands enormous capital against risk that is already hedged away.

Portfolio margin — risk-based

> Portfolio margin — a risk-based methodology that sets margin equal to the worst modelled loss of the *whole account* across a defined grid of price (and sometimes volatility) shocks. Longs, shorts, and options that offset each other net down, so a well-hedged book posts far less. Permitted for qualifying accounts under FINRA/CBOE rules.

> TIMS — the Theoretical Intermarket Margin System, the OCC's portfolio-margin engine for equities and equity options. It revalues the portfolio across roughly ±15% (equities) shock points — wider for narrow-based indices and single names — and takes margin as the largest loss in the grid.

WORKED EXAMPLE

Reg T vs portfolio margin on the SAME hedged book. A fund holds +$100mm long in Stock A and −$90mm short in closely correlated Stock B (a relative-value pair, beta ≈ 1).

Route 1 — Reg T. Each leg stands alone.
Long initial margin: $100mm × 50% = $50.0mm.
Short initial margin: $90mm × 50% = $45.0mm.
Total Reg T margin = $95.0mm.

Route 2 — portfolio margin (TIMS). Apply a ±15% shock to the correlated pair. In the worst modelled scenario the long loses and the short gains (or vice-versa), but because the legs are hedged the *net* loss is small. Say the worst-case net move leaves a residual directional and basis loss of about 15% on the $10mm net exposure plus a modelled basis/decorrelation charge — roughly $1.5mm + a $6.0mm basis add-on ≈ $7.5mm.
Total portfolio margin ≈ $7.5mm.

Capital freed. $95.0mm − $7.5mm = $87.5mm of margin released — over 12× more exposure per dollar of posted margin. That freed capital is why long/short and relative-value strategies live on portfolio margin.

House margin and house add-ons

> House margin / house add-on — a PB's own requirement layered on top of the regulatory or model minimum. The PB raises margin above the model for concentration, illiquidity, large size relative to average daily volume, or gap risk. A margin multiplier can double or triple the base model number.

The regime is only the starting point. Two funds on identical portfolio-margin models can face very different real requirements once house add-ons are applied, and in a stress the PB can raise add-ons intraday — the moment the fund can least afford it.

WORKED EXAMPLE

House add-on for a concentrated position. A fund holds a $60mm single-name long whose TIMS base margin is 15% = $9.0mm. The position equals 8× the stock's average daily volume, so the PB applies a concentration multiplier of 2.5× on the portion above a 3× ADV threshold, plus a liquidity add-on.

Base: $9.0mm. House concentration add-on: an extra +$9.0mm. Liquidity gap add-on: +$3.0mm.
All-in house margin = $21.0mm versus a $9.0mm model number. The fund's effective leverage on this name is less than half what the raw model implied.

VaR-based margin

> VaR-based margin — margin set from a Value-at-Risk estimate of the portfolio — the loss not expected to be exceeded at a given confidence (e.g. 99%) over a close-out horizon, estimated from historical or Monte-Carlo scenarios. Common at larger PBs and increasingly the standard for cross-asset books because it captures correlations across the whole portfolio.

SPAN for futures

> SPAN — Standard Portfolio Analysis of Risk, the margin system CME and most futures clearing houses use. SPAN revalues each position across a grid of scan ranges — defined moves in price and volatility — and takes the worst loss, then grants inter-commodity spread credits where correlated contracts offset.

WORKED EXAMPLE

SPAN scan-range margin. A fund is long 20 E-mini S&P 500 futures. Index at 5,500, multiplier $50 → notion-
al per contract = 5,500 × $50 = $275,000; total notional = $5.50mm. SPAN sets the scan range at a 5% adverse price move.

Step 1 — scan loss per contract. $275,000 × 5% = $13,750.

Step 2 — total SPAN initial margin. $13,750 × 20 = $275,000.

That is a 5% margin on $5.50mm of notional — roughly 20× leverage before house add-ons, far more capital-efficient than Reg T equity margin, because futures margin is purely risk-based.

Inter-commodity credit. If the fund were also short a correlated contract (say Nasdaq futures), SPAN would grant a spread credit — reducing the combined margin below the sum of the two standalone scan losses, because the positions partly offset.

Cross-margining across products

Cross-margining nets risk across *different products* — for example an index future against the basket of underlying cash equities, or listed options against their deliverables. Where a PB or clearing house recognises the offset, the fund posts margin only on the small residual risk after the hedge. It is the most capital-efficient regime of all, and the hardest to obtain, since it requires the products to clear or be held where a single risk model can see both legs.

Why portfolio margin frees capital — and its risks

Risk-based margin is powerful precisely because it rewards hedging: the better hedged the book, the less capital it strands. But that efficiency carries two dangers the desk must respect.

- Procyclicality. Risk models key off volatility and correlation. When markets stress, volatility spikes and historical correlations break; the model demands sharply more margin exactly when assets are falling and liquidity is scarce — forcing de-leveraging into a declining market.
- Add-ons in stress. House concentration and liquidity add-ons expand when a position becomes crowded or illiquid, so the thinly-margined efficiency of calm markets can evaporate overnight. A book comfortable at 6× in a calm regime can be forced to 3× in a week.

BEST IN CLASS

Elite financing desks never run a book at its maximum permitted leverage. They model their portfolio margin under stressed correlation and volatility — not today's calm inputs — and hold a buffer of unencumbered liquidity sized to a model recalibration and a wave of house add-ons hitting at once. They map which positions would trigger concentration or liquidity multipliers *before* putting them on, negotiate caps on how fast a PB can raise add-ons, and treat portfolio margin's efficiency as a resource to be spent deliberately — not a licence to lever to the ceiling and be forced to sell at the bottom.

05 · REPO & SECURITIES FINANCING

Financing the Fixed-Income Book

For a fund holding bonds, the repo market is the financing engine — a secured cash loan collateralised by the bonds themselves. Repo lets a fund fund a Treasury position at a razor-thin haircut, earn the carry between the bond and the repo rate, and lever that carry to extraordinary multiples. It is also where the most spectacular de-levering events begin when haircuts widen.

Repo from the fund's side

> Repurchase agreement (repo) — a sale of securities with an agreement to buy them back later at a set higher price. Economically it is a secured cash loan: the fund sells its bond, receives cash, and repurchases it at maturity for the original price plus interest (the repo rate). From the fund's seat it is a way to *borrow cash* against a bond it owns.

> Reverse repo — the mirror trade from the cash lender's side: buy the security now, sell it back later, earn the repo rate. A fund with surplus cash uses a reverse repo to earn a secured return.

A fund funds a long bond in four steps: (1) buy the bond; (2) repo it out for cash; (3) use the cash to pay for the bond; (4) pay repo interest and keep the bond's coupon. The fund's own capital covers only the haircut. The return it keeps is the carry — the spread between the bond's yield and the repo rate it pays.

> Haircut (initial margin) — the discount the cash lender applies to the collateral. A 2% haircut on a $100mm bond means the lender advances only $98mm; the fund must fund the $2mm gap from its own capital. The haircut is the lender's buffer against the collateral falling in value before it can be sold.

> General collateral vs special — GC repo accepts any bond from a broad basket and trades near the general secured rate. A special is a specific bond in high borrow demand; cash lenders accept a *lower* repo rate to get it, so a special finances cheaply (even below zero), rewarding the holder.

> Term vs open/overnight — overnight repo rolls each day; open repo continues until either side cancels; term repo locks the rate and cash for a fixed period (e.g. 1 week to 3 months). Term protects against a funding squeeze; overnight is cheapest but must be rolled every day — the rollover risk.

WORKED EXAMPLE

Financing a $100mm Treasury position via repo. The fund owns $100mm (market value) of a Treasury note yielding 4.60%. It repos the bond at a 2% haircut and a repo rate of 5.00% (ACT/360), term 30 days.

Step 1 — cash raised. $100mm × (1 − 0.02) = $98.0mm. The fund funds the remaining $2.0mm from its own capital.

Step 2 — repo interest for 30 days. $98.0mm × 5.00% × (30/360) = $408,333.

Step 3 — coupon income for 30 days. $100mm × 4.60% × (30/360) = $383,333.

Step 4 — net carry (30 days). $383,333 − $408,333 = −$25,000. Here repo funding (5.00%) exceeds the bond yield (4.60%), so the *raw* carry is negative — the position only works if the trade captures price appreciation or a basis/spread convergence, not coupon carry. Flip the repo rate to 4.20% (a special) and carry turns positive: coupon $383,333 − interest $343,000 = +$40,333 per 30 days on $2mm of capital.

Sponsored repo and FICC central clearing

> Sponsored repo — a scheme in which a clearing member (the *sponsor*) brings a buy-side client into central clearing at the FICC (Fixed Income Clearing Corporation). The client trades cleared rather than bilateral; FICC becomes counterparty to both sides, netting exposures and cutting the dealer's balance-sheet use — which often lowers the rate the fund pays.

US regulators are pushing an ever-larger share of Treasury repo into central clearing. For funds, sponsored access means cheaper, more robust funding and less reliance on any single dealer's balance sheet — but it concentrates exposure in the CCP and its default waterfall.

Tri-party vs bilateral

In bilateral repo the two parties settle collateral directly between their own accounts, bearing all the operational and valuation work. In tri-party repo a neutral agent (BNY, JPM, Euroclear, Clearstream) holds the collateral, values it daily, applies haircuts and eligibility, and swaps collateral as needed. Tri-party dominates GC funding for its operational ease; bilateral and cleared dominate specials and basis trades where precise collateral control matters.

The Treasury cash-futures basis trade

> Treasury cash-futures basis trade — a relative-value trade that buys a cash Treasury (funded in repo) and sells the corresponding Treasury future, capturing the small price gap (the "basis") that converges as the future approaches delivery. Because the gap is tiny, funds lever it enormously — commonly 50× to 100× — financing the entire cash leg in repo at a low haircut.

WORKED EXAMPLE

The basis trade — funding, leverage, and a haircut shock. A fund puts on $1,000mm of the cash Treasury leg, funded in repo at a 2% haircut.

Step 1 — capital at risk. Cash raised = $1,000mm × 98% = $980mm. The fund posts only the $20mm haircut from capital → 50× leverage on the cash leg ($1,000mm ÷ $20mm).

Step 2 — the edge. Suppose the basis is worth ~8bps of the notional over the trade life: $1,000mm × 0.08% = $0.8mm — small in absolute terms, but $0.8mm on $20mm of capital is a 4% gross return before financing, magnified by the 50× frame.

Step 3 — the haircut-widening shock. In a stress the repo lender raises the haircut from 2% to 5%. Required capital jumps from $20mm to $1,000mm × 5% = $50mm — a $30mm top-up demanded immediately. If the fund cannot post it, it must sell cash Treasuries into a falling market, pressuring the basis further and forcing more funds to de-lever in the same direction. This is the self-reinforcing unwind regulators fear in the basis trade.

Roll and term-out considerations

Overnight repo is cheapest but exposes the fund to a rollover failure every morning. The desk manages this by terming out a portion of the funding — locking 1-week to 3-month repo so a sudden GC rate spike or haircut widening cannot halt the book overnight — and by laddering maturities so only a slice rolls on any given day. The trade-off is cost: term repo prices above overnight, so the desk pays an insurance premium for funding stability.

BEST IN CLASS

The best repo desks never fund a levered basis book entirely overnight. They term out and ladder the core of the position so no single day's roll can break them, hold a haircut-widening buffer of unencumbered cash sized to a stress move (2% → 5%+), and diversify across cleared/sponsored FICC and multiple bilateral counterparties so one dealer pulling back cannot force a fire-sale. They size the trade to survive the haircut shock, not merely to maximise the carry in calm markets.

06 · SYNTHETIC FINANCING

Swaps, TRS & Delta-One

Sometimes a fund wants the economic return of an asset without owning it. Through a total return swap, the dealer holds the asset and the fund holds a contract that passes it the full return against a financing leg. Synthetic financing is balance-sheet-light, operationally simple, and opens markets the fund cannot reach directly — and it is the quiet engine behind much of the industry's leverage.

Total return swaps and CFDs

> Total return swap (TRS) — a bilateral contract in which the dealer buys and holds a reference asset and pays the fund the total return — price moves plus dividends/coupons — while the fund pays a financing leg, usually SOFR + spread on the full notional. The fund never owns the asset; it holds only the swap.

> Contract for difference (CFD) — a closely related bilateral contract, common outside the US, that settles the price difference on the reference asset in cash. Economically similar to a TRS on a single name.

> Delta-one desk — the dealer desk that provides these products. "Delta-one" means the instrument tracks the underlying one-for-one (delta ≈ 1, no optionality). The desk hedges by holding the physical asset, so it is long exactly what the fund is synthetically long.

The economics vs cash prime brokerage

In cash prime the fund owns the asset and borrows against it; the asset, the margin loan, and the collateral all sit on the fund's balance sheet. In a TRS the asset sits on the *dealer's* balance sheet and the fund carries only a contract. Both deliver leverage and the same economic exposure; they differ in balance-sheet treatment, margin, operational load, and sometimes tax.

- Synthetic long. Fund receives the asset's total return, pays SOFR + spread. Equivalent to owning the asset on a margin loan — without the asset on its books.
- Synthetic short. Fund pays the asset's total return, receives SOFR − spread. Equivalent to borrowing and shorting the stock — without arranging a physical borrow.

Why funds use swaps

- Balance-sheet / leverage efficiency. The asset is off the fund's books, easing gross-leverage limits and often the margin required.

- Market access. Swaps reach restricted or hard-to-access markets (e.g. certain onshore emerging markets) where the dealer holds the local access and the fund takes the return synthetically.
- Tax and withholding. A swap can change the dividend-withholding treatment relative to direct ownership, a major driver in cross-border equity financing.
- Operational simplicity. One contract replaces custody, settlement, corporate-action processing, and physical stock borrow.
- Confidentiality. The fund's position is not visible as a holding — a double-edged feature, as Archegos showed, since dealers may not see the fund's aggregate exposure across counterparties.

Financing-leg mechanics, resets, and give-up

The financing leg accrues on the swap's notional at SOFR + spread and settles on periodic reset dates (often monthly). At each reset the notional is marked to the current asset price, the performance leg (price return plus income) is exchanged, financing is paid, and the notional re-strikes for the next period. Swaps may be bilateral or given up to a clearing house where eligible; clearing adds margin and a CCP counterparty in place of the bilateral dealer.

WORKED EXAMPLE

TRS financing-leg payment over one reset. A fund holds a synthetic long via a TRS on $80mm notional. Financing leg = SOFR + 35bps, SOFR = 5.00% → 5.35%, ACT/360. Reset period = 30 days. Over the period the asset rises 1.20% and pays no dividend.

Step 1 — financing paid by the fund. $80mm × 5.35% × (30/360) = $356,667.

Step 2 — performance received by the fund. $80mm × 1.20% = $960,000.

Step 3 — net exchanged at reset. $960,000 − $356,667 = +$603,333 to the fund. The notional then re-strikes at the new, higher price for the next period.

WORKED EXAMPLE

Cash margin loan vs TRS — which wins. A fund wants $100mm of long exposure for one year. SOFR = 5.00%.

Route 1 — cash margin loan. Fund owns the shares; posts $15mm equity under portfolio margin, borrows $85mm at SOFR + 50bps = 5.50%.

Component	Calculation	Annual $
Margin-loan interest	$85mm × 5.50%	(4.675mm)
Opportunity cost on $15mm margin	$15mm × 5.00%	(0.750mm)
All-in financing cost		(5.425mm)

Route 2 — total return swap. Dealer holds the $100mm; fund pays SOFR + 35bps = 5.35% on full notional, posts $10mm initial margin.

Component	Calculation	Annual $
Financing leg	$100mm × 5.35%	(5.350mm)
Opportunity cost on $10mm margin	$10mm × 5.00%	(0.500mm)
All-in financing cost		(5.850mm)

Read the result. Cash costs $5.425mm; the TRS costs $5.850mm — the cash loan is cheaper by $0.425mm (≈43bps on the exposure). But the TRS ties up $5mm less margin ($10mm vs $15mm) and keeps the position off the balance sheet. When the TRS spread tightens below the margin-loan spread — or when balance-sheet, access, or withholding benefits dominate — the swap wins. This is the daily financing decision.

WORKED EXAMPLE

Synthetic short vs borrowing the stock. The fund wants $50mm of short exposure to a name with a 250bps physical borrow fee.

Physical short. Borrow the stock, post cash, receive rebate = SOFR − 250bps = 2.50%; the borrow fee is the 250bps drag → cost $50mm × 2.50% = $1.25mm/yr (versus holding cash at SOFR).

Synthetic short (TRS). Fund pays the asset's total return and receives SOFR − spread. The dealer embeds the same scarcity: it quotes the short-financing leg at SOFR − 250bps, so the fund's net financing cost is economically the same ~250bps — but it avoids arranging the locate, recall risk, and buy-in exposure, and keeps the short off its books. When the dealer can source the borrow more cheaply than the fund (scale, inventory), the synthetic route can even beat the physical fee.

BEST IN CLASS

Sophisticated desks price cash and synthetic routes side-by-side on every core position and let the all-in cost, margin use, balance-sheet relief, access, and withholding treatment decide — not habit. They lock term financing legs on swaps so a dealer cannot reprice the spread in a stress, diversify synthetic exposure across multiple delta-one desks so no counterparty holds a concentrated, invisible view of the book, and — the clearest post-Archegos lesson — they aggregate their own synthetic exposure across all dealers internally, never assuming confidentiality from the dealers relieves them of managing their true, consolidated leverage.

07 · DERIVATIVES MARGIN

Cleared & Uncleared Margin

Every derivative a fund trades carries a margin footprint. On the cleared side a central counterparty sets the number through a risk model; on the uncleared side a post-crisis rulebook forces two-way margin that must be locked away from reuse. This chapter builds the number from first principles, traces a full day of margin movement, and shows how portfolio and cross-margin programmes shrink the bill.

The two worlds of margin

A fund reaches the derivatives market through two doors. The first is central clearing, where a clearing house stands between every buyer and seller. The second is the bilateral world, where the fund faces a dealer directly under an ISDA Master Agreement and a Credit Support Annex. The margin mechanics differ sharply between the two, and a modern financing desk must master both because a single macro trade often spans them — a cleared interest-rate swap hedged with an uncleared swaption, say, or an exchange-traded future offset against a bespoke basket.

> Central counterparty (CCP) — a clearing house that interposes itself between the two sides of a trade by novation, becoming buyer to every seller and seller to every buyer. It guarantees performance and collects margin from every clearing member. Examples: CME, LCH, ICE Clear, Eurex.

> Futures commission merchant (FCM) — the clearing member through which a fund accesses a CCP. The fund is not a direct member; it faces the FCM, which faces the CCP. The FCM passes through CCP margin and almost always adds its own house margin on top.

> Novation — the legal replacement of one contract (fund vs dealer) with two contracts (fund vs CCP, dealer vs CCP). After novation the original counterparties no longer face each other; both face the clearing house.

Cleared margin: the CCP's four demands

A CCP protects itself with a layered defence. A fund that clears through an FCM encounters four distinct charges, and conflating them is a classic rookie error.

- Initial margin (IM). A buffer sized to cover the worst plausible loss over the close-out period — the days the CCP needs to hedge or auction a defaulter's book (typically one to two days for liquid futures, five days for cleared swaps). It is posted up front and held; it is not netted against daily moves.

- Variation margin (VM). The daily settlement of the mark-to-market move. The loser pays the winner in cash, every day, in full. For listed futures VM is literally paid away and realised (settlement-to-market); for many cleared swaps it is collateral (collateral-to-market). This distinction changes the funding cost of the position.
- Default-fund contribution. A mutualised pool each member funds. If a defaulter's own IM and default-fund slice are exhausted, surviving members' contributions absorb the rest. This is the member's exposure to *other* members' failures.
- Add-ons. Extra IM for concentration, liquidity, wrong-way risk, and large positions relative to market depth. The FCM then stacks its own house multiplier above the CCP figure.

> SPAN — Standard Portfolio Analysis of Risk, the scenario-grid IM engine long used by CME and others. It revalues a portfolio across a fixed array of price and volatility shifts (the "risk array"), takes the worst scenario loss, then applies inter-month and inter-commodity spread credits plus a short-option minimum. CME's successor, SPAN 2, blends this with a filtered historical VaR.

> VaR-based IM — most cleared-swap CCPs (LCH SwapClear, CME IRS) size IM with a historical or expected-shortfall Value-at-Risk model over a 5–10 year look-back plus a stressed period, scaled to the margin period of risk and a high confidence level (99%+).

WORKED EXAMPLE

CCP initial margin on a Treasury-future position via a SPAN-style scan.

A macro fund is long 500 contracts of the 10-year Treasury note future. Contract size is $100,000 face; the future trades at a price of 112-00, so notional ≈ 500 × $112,000 = $56.0m.

Step 1 — scan range. The CCP sets the price scan range at 1.5 points per contract (its estimate of a one-day 99% move). One point = $1,000 per contract, so the per-contract scan loss = 1.5 × $1,000 = $1,500.

Step 2 — worst scenario. Across the 16-scenario risk array the worst loss for an outright long is the full down-shock: 500 × $1,500 = $750,000.

Step 3 — spread credits. The position is a naked outright, so there are no inter-month or inter-commodity offsets. Scan risk stands at $750,000.

Step 4 — CCP IM. Base IM = $750,000 (≈ 1.34% of notional).

Step 5 — FCM house add-on. The FCM applies a 1.25× multiplier for a concentrated directional book: 1.25 × $750,000 = $937,500.

The fund funds $937,500 of IM to carry $56m of exposure — leverage of roughly 60×, which is exactly why futures are capital-light and why a modest widening of the scan range (see Chapter 09) hurts so much.

Uncleared margin under UMR

Not every derivative can clear. Bespoke swaptions, exotic options, physically-settled FX, and single-name instruments stay bilateral. After 2008 the G20 ordered margin on these too. The result is the Uncleared Margin Rules.

> UMR — the Uncleared Margin Rules require counterparties to exchange two-way variation margin and, once in scope, two-way regulatory initial margin on non-centrally-cleared OTC derivatives. The rules phased in by size from 2016 to 2022.

> AANA — Aggregate Average Notional Amount, the group-level measure that determines UMR scope. A firm computes the average month-end gross notional of in-scope uncleared derivatives over a reference period (in the US, June–August). The six phases captured firms as the threshold fell; Phase 5 (Sept 2021) caught firms above ~$50bn AANA and Phase 6 (Sept 2022) caught firms above the final ~$8bn threshold, pulling in many mid-sized hedge funds for the first time.

> Regulatory IM (reg-IM) — UMR initial margin. It is two-way, must be segregated at a third-party custodian, and critically may not be rehypothecated — the receiver cannot reuse it. There is a $50m IM threshold per counterparty group below which no exchange is required, which keeps smaller relationships out of scope even once a firm is in-phase.

> SIMM — the ISDA Standard Initial Margin Model, the near-universal method for computing reg-IM. It is a sensitivity-based model: the firm computes the trade's delta, vega, and curvature to each risk factor, multiplies by prescribed risk weights, aggregates within risk classes using prescribed correlations, then combines risk classes. A shared model means both sides compute the same number and disputes shrink.

WORKED EXAMPLE

SIMM reg-IM on an uncleared interest-rate swap (delta component).
A fund holds a $200m 10-year receive-fixed USD interest-rate swap that is uncleared (it is embedded in a structured package). We size the interest-rate delta margin, which dominates IM for a plain swap.
Step 1 — sensitivity. The swap's PV01 (dollar value of a 1bp move) is about $190,000 for $200m of 10-year risk. SIMM expresses risk as sensitivity per basis point to each tenor bucket; concentrate it at the 10y vertex: s = $190,000 per bp.
Step 2 — risk weight. SIMM's risk weight for the 10y vertex of a regular-volatility currency (USD) is in the region of 53 (basis-point units). Weighted sensitivity WS = s × RW = $190,000 × 53 = $10.07m.
Step 3 — concentration threshold. SIMM caps single-vertex concentration; here the position sits below the USD concentration threshold, so no scaling applies.
Step 4 — aggregate. With risk at a single vertex and one currency, the interest-rate delta margin K ≈ the weighted sensitivity = ≈ $10.1m. A real swap spreads risk across tenors and the correlation matrix nets it down, but the order of magnitude — roughly 5% of notional for a 10-year swap — is the number a treasurer carries in their head.
Step 5 — segregate. That $10.1m of reg-IM moves to a third-party custodian (e.g. BNY, Euroclear, Clearstream) into a segregated account the dealer cannot touch or reuse. It is dead weight on the fund's liquidity — the price of the 2008 lesson.

The daily margin cycle and portfolio margining

Margin is not a one-off. It is a daily operational heartbeat: start-of-day positions are revalued, VM is computed and called, IM is recalculated as positions and volatility change, disputes are raised against an MTA, and collateral is delivered intraday. Across cleared products, a fund can lower the total through cross-margining and portfolio-margin programmes — for example CME/FICC cross-margining of Treasury futures against cleared repo, which lets offsetting rate risk net before IM is struck.

WORKED EXAMPLE

A daily VM flow with a mark move.
On Monday the fund is long 500 Treasury-note futures at a settlement price of 112-00. On Tuesday the future settles at 111-16 (111.50), a fall of 0.50 points.
VM = price change × point value × contracts = −0.50 × $1,000 × 500 = −$250,000.
Because listed futures settle-to-market, the fund pays $250,000 of cash VM to the CCP via its FCM that evening; the loss is realised and the contract is rebooked at 111.50. If instead this were a collateral-to-market cleared swap, the fund would post $250,000 of eligible collateral that the CCP holds but does not realise. The funding distinction is real: settled VM is a cash outflow the treasury must have ready at start of day, which is why the start-of-day cash position (Chapter 08) is sacred.

BEST IN CLASS

A best-in-class desk never treats margin as a number it is told. It pre-computes IM intraday with its own SPAN and SIMM replicas, reconciles the FCM/CCP figure to the penny every morning, and knows before the open what a trade will add to IM at each venue. It routes risk to the venue where the margin is cheapest after netting (cleared cross-margin vs bilateral SIMM), and it treats segregated reg-IM as locked, unavailable liquidity in every stress model — never as a reserve it can tap.

08 · MULTI-PRIME & TREASURY OPS

Multi-Prime, Counterparty Risk & Treasury Operations

A single prime broker is cheap, simple, and fragile. A world-class fund spreads its book across several primes to diversify counterparty risk, compete financing, and secure borrow — then runs a treasury-operations backbone precise enough to know its true net exposure to each one every morning. This chapter shows how to allocate across primes, measure real counterparty exposure net of collateral, sweep idle cash, and run the daily plumbing.

Why funds run multiple prime brokers

The 2008 failure of Lehman Brothers taught the buy side a permanent lesson: your prime broker holds your assets and finances your book, so its health is your risk. A fund that cleared and custodied everything at Lehman found its rehypothecated collateral trapped in the bankruptcy estate for years. From that point, multi-prime became standard practice. There are four distinct reasons to run more than one.

- Counterparty diversification. No single failure or de-risking event can freeze the entire book. If one prime pulls financing or fails, the others keep the fund trading.
- Financing competition. Primes compete on margin-loan spreads, borrow rates, and swap financing. Running two or three live relationships keeps every quote honest. The spread saved is pure alpha.
- Capacity. A large fund can exceed a single prime's balance-sheet appetite or concentration limits. Spreading the book keeps each relationship inside its comfort zone.
- Borrow access. Different primes have different securities-lending inventories. For a short book, access to a hard-to-borrow name at one prime can be the difference between expressing a thesis and abandoning it.

> Rehypothecation — a prime broker's reuse of a client's collateral or long securities to fund its own activities. Reused assets are not protected the same way segregated assets are; in a prime's default they can become a general unsecured claim. US rules (Reg T / SEC 15c3-3) cap customer rehypothecation at 140% of the debit balance, but excess margin left with a prime is the exposure to watch.

> Asset segregation — holding client assets in accounts legally separated from the prime's own balance sheet, so they are protected and returnable in a default. Funds negotiate to keep excess margin and unencumbered longs in segregated or custody accounts rather than leaving them available for reuse.

Allocating the book across primes

Allocation is a constrained optimisation. The desk routes each position to the prime that finances it cheapest, subject to counterparty-concentration caps, borrow availability, and the need to keep each relationship

economically meaningful (primes drop clients who send only the unprofitable flow). A common discipline: cap any single prime at, say, 40% of financed assets, keep at least two primes able to carry the whole book at a pinch, and concentrate hard-to-borrow shorts where the inventory lives.

WORKED EXAMPLE

Net counterparty exposure to a prime broker.
A fund runs the following at Prime A:

Item	Amount	Direction of exposure
Long securities (financed, held in custody)	$800m	Fund's asset at the prime
Margin loan drawn against the longs	$560m	Fund owes the prime
Short sale proceeds held by prime	$300m	Fund's cash at the prime
Excess margin / free cash left at prime	$120m	Fund's cash at the prime
Rehypothecation permitted on longs	up to 140% of $560m = $784m	At-risk if reused

Step 1 — gross assets at the prime. Longs held $800m + short proceeds $300m + free cash $120m = $1,220m of the fund's property sits at Prime A.
Step 2 — net of what the fund owes. The margin loan of $560m is the prime's claim on the fund. Net fund property at risk = $1,220m − $560m = $660m.
Step 3 — the real exposure. In a prime default the margin loan nets against the longs via the prime brokerage agreement, but the $120m of excess cash and any *rehypothecated* longs are the dangerous slice. If the prime has rehypothecated the full permitted $784m, the fund's recovery on reused assets ranks as an unsecured claim.
Lesson. The headline "$800m of longs" is not the exposure. The exposure is the ~$660m of net property — and above all the $120m of free cash plus any rehypothecated collateral that could be trapped. The fix: sweep the $120m out nightly (below) and cap rehypothecation contractually.

Cash management and sweeps

Idle cash left with a prime earns a poor rate and sits as counterparty exposure. Treasury sweeps it every night into instruments that are safer, higher-yielding, and off the prime's balance sheet.

> Cash sweep — the automated nightly movement of excess cash out of the prime-brokerage account into money-market funds, reverse-repo, or Treasury bills, reducing both counterparty exposure and cash drag.

> Reverse repo — the fund lends cash and takes high-quality securities (Treasuries) as collateral overnight, earning the repo rate with near-zero credit risk. The mirror image of the repo financing a fund uses to fund longs.

WORKED EXAMPLE

Yield pickup from sweeping idle cash.
The fund has $120m of free cash. Benchmark overnight SOFR = 5.00%.

Home for the cash	Rate earned	Annual income
Left with the prime (uninvested credit)	SOFR − 75bp = 4.25%	$5.10m
Government money-market fund	SOFR − 8bp = 4.92%	$5.90m
Overnight reverse repo (Treasury collateral)	SOFR − 2bp = 4.98%	$5.976m
Pickup: reverse repo vs leaving at prime	+73bp	+$0.876m / year

Sweeping $120m into reverse repo earns an extra $876,000 a year and simultaneously removes that cash as unsecured exposure to the prime — the collateral is Treasuries the fund now holds. Two benefits, one trade. At a $5bn fund, disciplined sweeping of all idle balances routinely adds several million dollars of return a year for no incremental risk.

Collateral management and optimisation

A fund posts margin at CCPs, FCMs, primes, and bilateral dealers. The discipline is to meet every call with the cheapest-eligible asset and keep scarce, high-value collateral free. Where a counterparty demands HQLA but the fund holds corporates or equities, an upgrade trade (repo/sec-lending) swaps weaker collateral for government bonds — at a spread cost and with rollover risk. A downgrade does the reverse to free up HQLA when it is scarce.

> Cheapest-eligible collateral — for each margin obligation, post the asset with the lowest all-in opportunity cost that still satisfies the eligibility schedule and haircut, keeping high-yielding or scarce assets working elsewhere.

The treasury-operations backbone

None of this works without a disciplined operations layer. The daily cycle a fund treasury runs:

- Start-of-day cash. A single, reconciled view of cash and collateral across every prime and custodian before the market opens. This is the foundation; everything else assumes it is right.

- Daily margin reconciliation. Recompute each prime's and CCP's margin independently, compare to their call, and dispute differences against the MTA. A surprise margin call that turns out to be a booking error must be caught the same morning.
- Interest and financing claims. Verify every margin-loan accrual, rebate, borrow fee, and swap-financing coupon. Primes make errors; the fund that checks recovers real money.
- Fails management. Chase failed settlements. A fail ties up cash or exposes the fund to buy-in risk and, under CSDR in Europe, cash penalties.
- Standing settlement instructions (SSIs). Maintain correct, pre-agreed delivery instructions per counterparty and currency so settlement is straight-through and not manual.

BEST IN CLASS

A best-in-class fund treats operations as a control, not a cost. It reconciles margin and cash to the penny across all primes by the open, sweeps every idle balance nightly into reverse repo or government MMFs, measures net (not gross) counterparty exposure daily and caps it per prime, keeps at least two primes able to carry the full book, and contractually limits rehypothecation while segregating excess margin. The cash desk and the collateral desk share one real-time inventory, so the cheapest-eligible asset is posted automatically and no high-value collateral is ever wasted on a call that plain cash could have met.

09 · LIQUIDITY RISK & MARGIN STRESS

Liquidity Risk & Margin Stress

A fund does not fail because its thesis is wrong; it fails because it runs out of cash before the thesis pays off. Liquidity risk is the gap between what the fund can turn into cash quickly and what it may be forced to pay quickly. This chapter separates the two liquidities, maps the fund's unique liability side, stress-tests margin under a volatility spike, and sizes the cash buffer that keeps the fund alive.

The two liquidities

> Asset (market) liquidity — how fast and how cheaply a position can be sold for cash without moving the price. Treasuries are highly liquid in calm markets; a large stake in an illiquid small-cap, a private credit position, or a complex structured note is not.

> Funding liquidity — the fund's ability to meet cash obligations as they fall due: VM calls, IM increases, margin-loan rolls, redemptions, and borrow-fee payments. A fund can be solvent on paper yet fail if funding liquidity dries up.

The two interact viciously. In a stress, asset liquidity evaporates (buyers vanish, haircuts widen) at precisely the moment funding liquidity is most needed (margin calls spike). March 2020 is the canonical case: even US Treasuries became hard to sell at a fair price exactly when everyone needed cash. "Risk-free" and "always liquid" are not the same thing.

The liability side unique to funds

A bank funds with deposits; a fund funds with investor capital that can be withdrawn under defined terms. Those terms are the fund's liability-side liquidity, and a mismatch against asset liquidity is the root of most blow-ups.

- Lock-up — a period after investment during which capital cannot be redeemed at all (commonly one to three years for less-liquid strategies).
- Redemption notice period — the days' notice an investor must give before a redemption date (e.g. 90 days), giving treasury time to raise cash.
- Gate — a cap on the percentage of fund NAV that may be redeemed on any one date (e.g. 25%), slowing an exodus so the fund is not forced to dump assets.
- Side pocket — an account that ring-fences illiquid positions so redeeming investors cannot force their fire-sale; they are paid out only as those assets are realised.

> Asset-liability liquidity mismatch — the gap between how fast the fund's assets can be sold and how fast its capital can be withdrawn. A fund offering monthly liquidity against a book that takes months to liquidate is running a mismatch that only investor-side tools (locks, gates, side pockets) can bridge.

Margin stress testing

IM is not static. CCPs and SIMM models are driven by volatility and haircuts; when markets move, required margin jumps — and the fund must find the extra cash the same day. The discipline is to shock the book and read off the incremental call *before* it happens.

WORKED EXAMPLE

Margin stress under a volatility spike and haircut widening.
A relative-value fund runs a $5.0bn long Treasury book financed in repo, plus cleared rate futures.
Repo leg — haircut widening. In calm markets the repo haircut on Treasuries is 2%, so on $5.0bn of bonds the fund posts $100m and borrows $4.9bn. In the stress, dealers widen the Treasury haircut from 2% to 5%.
New margin required = 5% × $5.0bn = $250m. Incremental cash to find = $250m − $100m = $150m.
Futures leg — vol spike. The CCP widens the price scan range as realised volatility doubles. Base IM on the futures was $90m; the CCP roughly doubles the scan, lifting IM to $180m. Incremental cash = $90m.
VM leg — the mark move. Yields gap 30bp against the book on day one. On $5.0bn of 10-year-equivalent risk (PV01 ≈ $4.6m per bp), the one-day loss ≈ 30 × $4.6m = $138m of VM paid away in cash.

Source of call	Driver	Extra cash
Repo haircut widening	2% → 5%	$150m
Futures IM	Vol doubles, scan widens	$90m
Variation margin	30bp adverse gap	$138m
Total cash needed in one day		$378m

A single bad day demands $378m of fresh cash. If the fund's unencumbered liquidity is below that, it is a forced seller into a falling market — the LDI and basis-trade mechanism of Chapter 10.

Rollover risk and terming-out financing

> Rollover (roll) risk — the risk that short-term financing (overnight or term repo, a margin loan, a collateral-upgrade trade) cannot be renewed on the old terms, or at all, when it matures. Lehman and every funding run begins here.

The defence is to term out financing: lock repo and margin-loan tenor further out (30/60/90-day term repo, evergreen or notice-period prime financing) so that no single day can see the whole book come up for renewal. Terming out costs a spread over overnight funding — that spread is the insurance premium against a run. A desk also staggers maturities so only a slice rolls on any given day.

The liquidity ladder and buffer sizing

> Days-to-liquidate — the time to convert the book to cash without undue price impact, usually computed position-by-position as the fraction of average daily volume the fund can sell per day, then rolled up.

> Redemption coverage — the ratio of liquidity available over a horizon to the cash that could be demanded (potential redemptions after gates/notice + stressed margin calls) over the same horizon. Above 1.0× the fund survives; below it, the fund is a forced seller.

WORKED EXAMPLE

A liquidity-coverage check over 30 days.

Sources of cash (30 days)	Amount
Unencumbered cash + government MMFs	$300m
Liquidation proceeds, liquid sleeve (haircut-adjusted)	$450m
Total available liquidity	$750m

Potential uses of cash (30 days)	Amount
Redemptions after notice + 25% gate	$400m
Stressed margin calls (from the stress above, scaled)	$378m
Total potential outflow	$778m

Coverage ratio = $750m ÷ $778m = 0.96×. Below 1.0×. The fund is $28m short in a combined redemption-and-margin stress. The treasurer must act now: term out more repo to cut the margin spike, raise the cash buffer, tighten the gate, or shed the least-liquid positions. A best-in-class desk runs this check daily and targets coverage of at least 1.2× under its worst-case scenario, so a 0.96× reading is a flashing red light, not an academic number.

BEST IN CLASS

A best-in-class liquidity framework runs a daily coverage ratio that nets stressed margin calls and gated redemptions against unencumbered cash and haircut-adjusted liquidation proceeds, targeting ≥1.2× in the worst modelled scenario. It sizes the cash buffer to survive a multi-day shock — not an average day — treats segregated IM and encumbered collateral as unavailable, terms out and staggers financing so no single day rolls the whole book, and matches redemption terms (locks, notice, gates, side pockets) to the true days-to-liquidate of the book so the asset-liability mismatch never forces a fire sale.

10 · WAR STORIES

Archegos, LTCM, March 2020, LDI

Every rule in this guide was written in blood. Four financing-desk post-mortems — a concentrated swap book, a convergence fund, a Treasury basis unwind, and a leveraged pension hedge — show the same machinery failing in different costumes. For each: the mechanism that broke, the figures, and the explicit lesson for a fund treasurer.

Archegos (2021) — hidden leverage through total return swaps

Archegos Capital Management, the family office of Bill Hwang, built enormous concentrated positions in a handful of names (ViacomCBS, Discovery, GSX, Baidu) not by buying stock but through total return swaps at several prime brokers at once. Because the swaps were synthetic, each prime saw only its own slice — none saw the whole. Reported equity of roughly $10bn supported gross exposure estimated near $50–100bn: leverage on the order of 5× or more, concentrated in illiquid single names.

Mechanism. When the underlying stocks fell in March 2021, the swaps generated VM calls across every prime simultaneously. Archegos could not meet them. The primes raced to close out — seizing and dumping the same concentrated stocks into a falling market. First movers (Goldman, Morgan Stanley) escaped; slow movers (Credit Suisse ~$5.5bn, Nomura ~$2.9bn) absorbed the losses. Total dealer losses were about $10bn.

> Hidden gross exposure — leverage that is invisible to any single counterparty because it is fragmented across many prime brokers via synthetic swaps. Each dealer underwrites its own exposure blind to the aggregate, so none sizes margin for the true concentration.

Lesson for a treasurer. Synthetic financing hides leverage from your counterparties — and from you if you are sloppy. Know your *aggregate* gross exposure across every prime, not the per-prime view each prime shows you. Concentrated single-name risk means a close-out crowds the exit; model what your own forced unwind does to the price. And on the sell side, the lesson is that TRS margin must reflect concentration and the possibility that other dealers hold the same book.

LTCM (1998) — convergence, leverage, and the liquidity spiral

Long-Term Capital Management ran convergence trades: long the cheap, less-liquid bond, short the rich, more-liquid one, betting spreads would narrow. Individually each trade was low-risk, so the fund levered hard — roughly 25× balance-sheet leverage ($~4.8bn equity against ~$125bn of assets) plus off-balance-sheet derivative notional reported above $1 trillion.

Mechanism. After Russia defaulted in August 1998, investors fled to liquidity. Spreads *widened* instead of converging — every LTCM trade lost at once because they shared the same hidden factor: a short position in liquidity. Losses triggered margin calls; raising cash meant selling into markets where LTCM *was* the market,

which widened spreads further and triggered more calls. The fund lost ~$4.6bn in weeks and was recapitalised by a Fed-brokered consortium of 14 banks (~$3.6bn).

> Liquidity death-spiral — leverage + mark-to-market margining + a one-way market where the fund is a dominant holder. Selling to meet calls moves the price against the remaining book, triggering more calls — a self-feeding loop. The same shape recurs in LDI (2022) and the basis trade (2020).

Lesson for a treasurer. Low volatility invites leverage; leverage manufactures the tail it pretends to avoid. Diversification is an illusion when every trade is secretly the same bet (here, long illiquidity). Size positions so your own liquidation does not move the market, and stress-test for correlations going to 1.0 in a panic.

March 2020 — the Treasury basis-trade unwind and dash for cash

The Treasury basis trade exploits the small gap between a cash Treasury and the matching futures contract: buy the cash bond, fund it in repo, short the future, and harvest the basis. The edge is tiny, so funds lever it 50–100× via repo. When the pandemic hit, everyone wanted cash; even Treasuries sold off as investors liquidated the safest, most liquid asset.

Mechanism. Falling Treasury prices produced VM calls on the cash legs; repo haircuts widened, forcing more margin; dealers, capacity-constrained, could not warehouse the flow. Funds unwound the basis trade all at once, selling cash Treasuries into a market with no bid, widening the dislocation. The world's deepest market seized. The Fed intervened on a massive scale — buying roughly $1 trillion of Treasuries in weeks and launching repo facilities — to restore function.

Lesson for a treasurer. "Risk-free" is not "always liquid." A highly-levered trade on a razor-thin spread cannot survive a haircut-widening and VM shock; size the buffer for the bad day. The Fed may backstop the system, but it will not backstop your fund — you must survive the gap on your own balance sheet. This episode drove the SEC's mandate to push Treasury repo into central clearing.

2022 UK LDI — leveraged gilt hedges and the collateral spiral

UK defined-benefit pensions used liability-driven investment: leveraged interest-rate and gilt exposure (via repo and swaps) to match very long liabilities on limited capital. The leverage meant a modest yield move produced a large collateral call.

Mechanism. The September 2022 "mini-budget" sent gilt yields up ~100–130bp in days. The leveraged hedges lost value, triggering huge VM calls. To raise cash, LDI funds sold gilts — pushing yields higher, triggering more calls: a textbook collateral spiral. 30-year gilt yields ran toward 5% and the market was days from disorderly failure. The Bank of England broke the loop with emergency gilt purchases (up to £65bn authorised) and a fast recapitalisation of the LDI pools.

Lesson for a treasurer. Leverage + daily margining + a one-way market is the same doom loop every time (LTCM, March 2020, LDI). Hold enough liquid, *eligible* collateral — not just liquid assets, but assets the

counterparty will accept without a punitive haircut — to survive a large, fast move. Pre-position collateral so you are not scrambling to transform it mid-crisis, when upgrade trades are most expensive and least available.

Episode	Leverage	What broke	Loss / scale
Archegos 2021	~5×+ via TRS	Hidden cross-prime concentration	~$10bn dealer losses
LTCM 1998	~25× + derivatives	Correlated convergence, liquidity spiral	~$4.6bn; Fed-brokered rescue
March 2020	50–100× basis trade	Treasury illiquidity, haircut + VM shock	Market seized; ~$1tn Fed buying
UK LDI 2022	~3–7× gilt hedges	Collateral spiral on yield spike	BoE up to £65bn backstop

WORKED EXAMPLE

The Archegos arithmetic a treasurer should have run.
Suppose $10bn of equity supports $50bn of gross single-name TRS exposure (5× leverage), concentrated so that one name is $12bn of the book.
Step 1 — the move. That name falls 20% in a day: mark loss = 20% × $12bn = $2.4bn on one position.
Step 2 — VM call. The dealers call $2.4bn of VM that evening across the primes holding it.
Step 3 — buffer check. If liquid buffer is, say, $1.5bn, the fund is $0.9bn short on day one from a single name — before the rest of the correlated book moves.
Step 4 — the spiral. Unmet calls trigger close-out; dealers dump the $12bn position, driving it down further, deepening losses on the rest. Equity of $10bn is wiped in days.
The lesson in numbers: at 5× leverage, a 20% move in a 24%-of-equity position is already larger than any reasonable buffer. Concentration, not leverage alone, is the killer.

BEST IN CLASS

A best-in-class desk reads these four as one lesson: measure aggregate gross exposure across all venues, never the fragmented per-prime view; assume correlations go to 1.0 and liquidity to zero in the stress; size positions so your own forced unwind does not move the price against you; term out and pre-position eligible collateral so you never transform in a crisis; and run the daily "what cash do I need on the worst day" check (Chapter 09) so the spiral never starts. The common failure is always the same — leverage plus daily margining plus a crowded exit — and the common defence is a liquidity buffer sized for the bad day, not the average one.

11 · LIQUIDITY CRUNCHES

A Century of Liquidity Crunches — Patterns & Calculations

Nearly every hedge fund that has died violently died the same way: not because its thesis was wrong, but because it ran out of cash while it was still right. This chapter catalogues the major liquidity crunches of the modern era from the leveraged investor's seat, extracts the repeating signatures, and works the arithmetic of margin spirals, two-sided runs and forced deleveraging so you can see your own book inside each one.

The universal anatomy of a liquidity crunch (fund's-eye view)

Strip the headlines away and every crunch runs the same seven-act play. What changes is the asset class, the acronym and the size of the hole — never the mechanics. Learn to feel the sequence from your own treasury desk and you will recognise act three while others are still congratulating themselves on act one.

Act 1 — The mismatch builds in calm times. Spreads are tight, vol is low, financing is cheap and plentiful. You lever up because the carry is thin and the only way to a double-digit return is gross exposure well above capital. Your liabilities (prime-broker financing, repo, investor capital with quarterly liquidity) are short and runnable; your assets (off-the-run bonds, structured credit, concentrated equity, convergence trades) are slower to sell. The gap between how fast you can be asked for money and how fast you can raise it is the crunch, pre-loaded and waiting.

Act 2 — A loss or vol shock hits. A default, a policy surprise, a de-peg, a crowded trade that reverses. The shock itself is usually survivable. What is not survivable is what the shock does to your financing.

Act 3 — The two-sided run. This is the act that kills. Your prime brokers raise haircuts and issue margin calls *at the same moment* your investors file redemptions. Funding runs from the asset side (lenders want more collateral) and the liability side (investors want their money) simultaneously. One alone is manageable; both together, against illiquid assets, is a vice.

Act 4 — Forced selling of the wrong things. To raise cash now, you sell what *can* be sold — your most liquid, highest-quality, most-loved positions — not the junk you would like to be rid of. The book gets worse, not better. Quality leaves, illiquidity stays.

Act 5 — The mark-to-market and margin spiral. Your selling pushes prices down; lower marks mean larger losses, lower equity and higher leverage; higher vol means larger haircuts and bigger margin calls; which forces more selling. The loop feeds itself.

Act 6 — Contagion through crowded trades and shared plumbing. You are not alone in the trade or at the prime broker. As peers deleverage the same positions, correlations you assumed were low snap to one, and the broker you share with the blow-up next door tightens terms on everyone.

Act 7 — Close-out or backstop. Either the prime broker cuts your financing and liquidates your collateral into the panic (close-out), or a backstop appears — a central-bank facility, a consortium, fresh capital — and the spiral breaks. Funds that reach act seven as a seller rarely reach act eight as a going concern.

> Fire sale — selling assets below fair value because you must raise cash immediately, not because you want to exit the position; the forced seller is a price-taker and the market knows it.

> Margin spiral — the self-reinforcing loop in which losses trigger margin calls, forced sales deepen losses, and the larger losses trigger still more margin calls.

> Haircut spiral — the collateral-side version: rising volatility raises haircuts, which forces deleveraging, which raises volatility, which raises haircuts again. Also called the "margin/haircut loop" or loss-spiral.

> Dash for cash — a system-wide flight in which even normally liquid, high-quality assets (e.g. Treasuries) are sold to raise cash, breaking the usual flight-to-quality pattern.

> Crowded trade — a position held by many leveraged investors at once, so that when one is forced to exit, all are hit together and diversification silently disappears.

> Redemption run — investors filing withdrawal notices en masse, forcing the fund to raise cash from the portfolio regardless of market conditions — a bank-run on the liability side.

> Forced deleveraging — cutting gross exposure not by choice but to satisfy a leverage ceiling, a margin call or a redemption, typically into a falling market.

> Prime-broker close-out — the prime broker's right, on a margin default, to seize and liquidate posted collateral and terminate financing — often all brokers at once, into the worst possible tape.

The repeating patterns — what every crunch has in common

Across a century of blow-ups the same handful of fingerprints recur. If you can only remember one thing from this chapter, remember this: it is almost never the thesis that kills a leveraged fund — it is the funding. LTCM's

convergence trades eventually converged. Archegos's names were not frauds. The positions were "right" on a long enough horizon; the fund simply did not own the horizon.

- Funding liquidity, not the thesis, is the cause of death. Solvency is about being right eventually; liquidity is about surviving until then. Crunches destroy the second, not the first.
- Short, runnable financing against slower assets. Overnight repo, daily-margined swaps and quarterly investor capital funding multi-month-to-liquidate positions. The maturity and liquidity transformation *is* the vulnerability.
- Hidden or synthetic leverage. Total-return swaps, options and futures let notional exposure dwarf posted capital. The leverage you cannot see on the balance sheet is the leverage that kills you (Archegos, LTCM's $1.25tn notional).
- Crowded, correlated positioning. Everyone in the same value/momentum factor, the same basis trade, the same EM convergence. In stress, correlations go to one and your "diversified" book behaves like a single leveraged bet.
- Collateral pro-cyclicality. Haircuts and initial margins are lowest in calm and spike in stress — they tighten exactly when you can least afford to post more. Margin is a short-vol position embedded in your funding.
- The two-sided run. Lenders and investors pull at once. Design your liquidity assuming they are correlated, because in a crunch they are.
- Asset–liability liquidity mismatch versus gates and notice. Survival often comes down to whether your redemption notice periods, gates and lock-ups are matched to how fast your assets actually sell.
- Concentration and single-counterparty dependence. One prime broker, one repo lender, one custodian — any single point of failure in the funding stack becomes a single point of death.

Signature	What it looks like in calm	What it does in the crunch
Liquidity mismatch	Cheap short financing, patient assets	Lenders refinance daily; you cannot sell daily
Hidden leverage	Notional >> capital via swaps/options	Small price move > entire equity
Crowding	Low measured correlation	Correlations → 1; everyone sells together
Pro-cyclical collateral	2% haircuts, thin margin	Haircuts double/triple; instant cash demand
Two-sided run	Stable financing + sticky capital	Margin calls + redemptions same week
Counterparty concentration	One PB, best terms	One pull = full close-out

A chronological field catalogue

Below is the leveraged investor's catalogue of the modern era's crunches. Chapter 10 already deep-dived LTCM (1998), Archegos (2021), the March-2020 dash-for-cash and the UK LDI episode (2022); the entries here add the others and, more importantly, surface the shared mechanics so the pattern — not the trivia — is what you carry away.

Year	Event	Leverage / mismatch	Trigger	Loss / run metric	Outcome
1987	Black Monday	Portfolio insurance = synthetic short futures; dynamic hedging	Equity wobble → programmatic selling	S&P −20.5% in one day	Feedback crash; Fed liquidity backstop
1994	Bond "massacre"	Levered carry & mortgage convexity books	Surprise Fed hikes (3% → 6%)	~$1.5tn global bond value lost	Askin, Granite funds fail; dealer losses
1997	Asian crisis	USD-funded EM carry; peg bets	Thai baht de-peg, contagion	Several EM FX −50%+	Capital flight; IMF programs
1998	LTCM / Russia	~26× balance sheet; ~$1.25tn notional	Russia default; flight to quality	~$4.6bn equity lost in weeks	$3.6bn dealer-consortium bail-in
2007	Quant quake (Aug)	Crowded, levered equity market-neutral factor books	One fund deleverages crowded factors	Multi-σ factor losses in 3 days	Sharp loss then snap-back; survivors de-levered
2008	Lehman collapse	Rehypothecated PB balances; frozen assets	Lehman bankruptcy (15 Sep)	Funds' assets trapped at LBIE for years	Rush to multi-prime; segregation reform
2008	Money-fund break / CP freeze	Funds reliant on CP & repo rollover	Reserve Primary "breaks the buck"	$3.45tn MMF outflows pressure	Fed/Treasury guarantee & facilities
2015	SNB CHF un-peg	Short-CHF carry; FX leverage 20–100×	SNB abandons 1.20 floor (15 Jan)	EUR/CHF −~30% in minutes	Everest sub-fund wiped; brokers blow up
2018	Volmageddon (Feb)	Short-vol ETPs; inverse-VIX leverage	VIX spike doubles overnight	XIV −96%, terminated	Short-vol complex collapses
2020	COVID dash-for-cash (Mar)	Treasury-basis trade ~50×+; levered relative value	Pandemic shock; margin + redemptions	Basis gaps; Treasuries sold off	Fed $ unlimited UST/ repo backstop
2021	Archegos	~5× via TRS across ≥7 unaware prime brokers	ViacomCBS drops; collateral shortfall	~$10bn+ dealer losses; fund wiped	Credit Suisse crippled; TRS scrutiny
2022	UK LDI	Pension liability hedges levered ~3–7× via repo/swaps	Gilt yields spike after mini-budget	Collateral calls > pooled fund NAV	BoE emergency gilt buying
2022	Crypto cascade	Terra reflexive peg; 3AC cross-margined leverage	UST/LUNA de-peg (May)	~$60bn LUNA; 3AC $3bn+ default	3AC liquidated; FTX fails (Nov)
2023	SVB / USDC depeg	Duration mismatch; uninsured deposit run	SVB solvency fear; $3.3bn USDC at SVB	USDC → $0.87; $42bn 1-day run	FDIC backstop; peg restored

The mechanics, as calculations

Patterns persuade; arithmetic proves. The following worked examples use illustrative figures grounded in the real events. They are the five or six calculations a financing officer should be able to do on a napkin, because in a crunch you will be doing exactly these sums with real money on the line.

WORKED EXAMPLE

1 — The fire-sale deleveraging multiplier. A fund runs 10× leverage: $10 equity against $100 of assets. A 5% adverse move hits the asset side.

```
Start:    Assets 100   Debt 90   Equity 10   Leverage = 100/10 = 10.0x
Loss:     5% of 100 = 5   →   Assets 95   Debt 90   Equity 5
New lev:  95 / 5 = 19.0x        (leverage nearly DOUBLED on a 5% move)

To restore 10x with equity now = 5:
   target assets = 10 x 5 = 50
   must sell = 95 - 50 = 45       (repay debt down to 45)
   sale as multiple of the $5 loss = 45 / 5 = 9x
```

A $5 loss forces roughly $45 of selling — nine times the loss — just to hold leverage constant. Every leveraged fund in the same assets is doing this at once, so the selling itself pushes prices down another leg, which is act 5: the spiral.

WORKED EXAMPLE

2 — The haircut / margin spiral (the March-2020 basis shape). Financing leverage is capped by the haircut: max leverage = 1 / haircut. A Treasury cash-futures basis book finances bonds in repo at a 2% haircut.

```
Calm:   haircut 2%  →  max leverage = 1/0.02 = 50x
        Book: $10,000mm bonds financed on $200mm margin (2% of 10,000)

Stress: haircut widens 2% → 5%
        Required margin = 5% x 10,000 = $500mm
        New cash demand  = 500 - 200 = $300mm  (2.5x the original margin)

If only $200mm cash is on hand, the shortfall of $300mm must come from
sales:
        bonds to sell to free $300mm margin at 5% haircut
        = 300 / 0.05 = $6,000mm of bonds dumped
```

A move in the *haircut alone* — no loss on the bonds yet — forces a $6bn fire-sale from a $10bn book. The selling cheapens the bonds, the dealer re-marks and re-haircuts, and the loop runs. This is the exact shape of the March-2020 Treasury-basis unwind that only stopped when the Fed became buyer of last resort.

WORKED EXAMPLE

3 — LTCM-style thin-equity wipeout. Early 1998: roughly $4.8bn equity on ~$125bn of balance-sheet assets, plus ~$1.25tn of derivative notional.

```
Balance-sheet leverage = 125,000 / 4,800 ≈ 26x

Equity-threatening move on the $125bn asset base:
   4% adverse = 0.04 x 125,000 = $5,000mm loss
   $5,000mm > $4,800mm equity   →   INSOLVENT on a ~4% move

Notional leverage = 1,250,000 / 4,800 ≈ 260x
   a few tens of bps across $1.25tn notional = billions
```

The convergence trades were built to profit as spreads narrowed. After Russia's August-1998 default, flight-to-quality made spreads *widen* — the "converging" legs diverged in lockstep across every market at once (correlations → 1), precisely the crowding pattern. With a ~4% buffer, LTCM did not need to be wrong for long; it needed the market to move 4% while it was still "right," and it did.

WORKED EXAMPLE

4 — The two-sided run (redemptions + margin at once). A fund has $1,000mm equity, runs 3× gross ($3,000mm of positions). In one quarter it faces a 25% redemption *and* a vol-driven margin increase.

```
Redemption:  25% x 1,000 = $250mm cash owed to investors
Margin rise: +$120mm additional collateral called by prime brokers
Total cash demand = 250 + 120 = $370mm, essentially same week

Cash on hand (buffer) = $80mm
Shortfall = 370 - 80 = $290mm  → must be raised by selling

To stay at 3x after paying out $250mm of equity:
   new equity ≈ 1,000 - 250 (paid) - losses; target gross = 3 x new equity
   so gross must fall from 3,000 toward ~2,250  →  ~$750mm of SALES
   plus the $290mm cash shortfall is met from the most liquid of those sales

With a 45-day notice period / soft gate:
   the $250mm redemption is DEFERRED past the stress window;
   immediate cash demand = 120 (margin) - 80 (buffer) = $40mm only
   forced sale ≈ $40mm vs ~$750mm — the gate converts a crisis into a
nuisance
```

Same portfolio, same shock. Without liability-management terms the fund dumps three-quarters of a billion into a falling market; with a 45-day notice and a modest buffer it posts $40mm and survives. Notice periods and gates are not legal boilerplate — they are liquidity capital.

WORKED EXAMPLE

5 — Days-to-liquidate under stress. A position is $600mm. The rule of thumb: never be more than, say, 20–25% of daily volume without severe impact. Normal daily volume lets you sell $150mm/day.

```
Calm:   days-to-liquidate = 600 / 150 = 4 days
Stress: volume HALVES  → tradable = 75/day
        impact DOUBLES → prudent participation halves again → ~37.5/day
        days-to-liquidate = 600 / 37.5 = 16 days

Redemption + margin horizon in stress:
   investors owed in T+3; margin due same-day/next-day
   you need cash in ≤ 3 days; the asset needs 16 days to clear cleanly

Structural liquidity gap = 16 - 3 = 13 days short
```

When the days-to-liquidate blows past the redemption-and-margin horizon, the fund is structurally short liquidity: it cannot meet its own calls at fair value and is condemned to sell at fire-sale impact. The fix is set before the crunch — hold liquid buffers, match notice to the longest days-to-liquidate, and stress volume/impact, not just price.

WORKED EXAMPLE

6 — Archegos-style concentrated synthetic leverage. Family office with ~$10bn of capital builds concentrated single-name exposure through total-return swaps spread across seven-plus prime brokers.

```
Suppose each of 7 PBs sees a book funded at ~5x, margined on ITS slice only:
   per-PB visible gross ≈ (10bn / 7) x 5 ≈ $7.1bn  "looks fine, 5x,
diversified"
   TRUE aggregate gross ≈ 10bn x 5 ≈ $50bn, concentrated in a handful of
names

Gap = each broker sees ~$7bn; the real position is ~$50bn in the SAME stocks
   When ViacomCBS fell ~30%:
   loss on the concentrated names overwhelmed the ~$10bn equity
   every PB called margin the same day; no one could see the others
   first-mover brokers closed out; the slow ones ate ~$10bn+ of losses
```

The mechanism is pure hidden-leverage-plus-concentration: notional dwarfed visible capital, TRS kept the positions off every broker's radar, and single-counterparty blindness meant no one priced the real crowding until the close-out began. The fund's seat lesson: leverage you spread to hide is leverage that detonates all at once.

The financing officer's early-warning dashboard

You cannot forecast the trigger, but you can watch the pressure gauges that always move before the pipe bursts. Monitor these daily; define a named, pre-agreed action for each red level so that in the crunch you execute a plan rather than improvise.

Indicator	Normal	Stressed	How to read it
Own weighted-avg haircut trend	Flat / falling	Rising across asset classes	Pro-cyclical margin turning against you; leverage ceiling dropping
Margin-to-equity	< 20–30%	> 50% and climbing	How much of your capital is already pledged; little left to meet new calls
Unencumbered cash vs 1-day stress call	> 2–3× cover	< 1× cover	Can you survive tomorrow's worst plausible call without selling?
FRA-OIS / cross-ccy basis	Tight, stable	Blowing wider	Bank funding stress and USD scarcity — financing about to get dear/scarce
Repo specialness / GC rates	Near policy rate	Spiking, bonds on special	Collateral scarcity; your repo rollovers may fail or re-price
Borrow fees on crowded shorts	Low, stable	Spiking / recall risk	Short-squeeze and crowding pressure; forced buy-ins loom
Redemption-notice pipeline	Net neutral	Net outflows accelerating	The liability-side run forming; size cash need for the next gate date
Counterparty CDS / AT1 spreads	Tight	Widening on your PBs	Your financier may be the next domino — ready a transfer plan
Days-to-liquidate vs redemption+margin horizon	DTL < horizon	DTL > horizon	The master gauge: if it inverts you are structurally short liquidity

> Liquidity is a coward — it is there when you do not need it and gone the moment you do. Treasury's job is to buy it while it is still brave.
>
> — trading-desk adage

BEST IN CLASS

An elite fund treasury does not try to predict the next crunch; it is built so that any crunch is survivable. Term out the financing: match funding tenor to asset liquidity, hold committed facilities and extend the shortest, most runnable liabilities. Run multi-prime with headroom: never depend on a single counterparty, keep spare margin and the ability to port positions, and know your rehypothecation and segregation terms cold. Carry a large unencumbered-cash buffer sized to the worst plausible one-day combined margin-plus-redemption call, not the average. Stress daily on both sides at once — margin and redemptions as correlated events — with named, pre-agreed triggers and a written de-risking playbook so the crunch is executed, not debated. Manage the liability like capital: notice periods, gates and lock-ups matched to the real days-to-liquidate of the book, so the two-sided run can be deferred past the storm. The funds that walk out of a crunch are rarely the ones who were most right; they are the ones who pre-paid for liquidity in the calm and never let a single lender, a single asset or a single day's volume become the thing that could kill them.

12 · CAPSTONE

What "Best in Class" Looks Like

Everything in this guide converges here: the target operating model for a world-class hedge-fund treasury and financing desk. Financing is run as a managed P&L, not a back-office cost; every position is routed to its cheapest venue through a financing waterfall; margin, counterparty exposure, and liquidity are watched in real time; and the individual who masters the vocabulary, the frameworks, and the canon becomes recognised as an expert. This is the destination.

Financing as a managed P&L

The first mental shift of a best-in-class desk is that financing is alpha. Two funds holding the identical trade keep different returns if one finances it 20bp cheaper. At scale that spread is pure, repeatable, uncorrelated alpha — so the financing desk is a profit centre with its own P&L, its own targets, and its own daily dashboard. The weighted-average financing spread the fund pays is a number the CIO watches, because every basis point shaved is return delivered with no new market risk.

> Financing P&L — the running tally of what the treasury desk earns or saves against a benchmark: spread compression on margin loans and swaps, sweep income, borrow-fee recoveries, and collateral-optimisation savings, net of the cost of terming-out insurance. Run as a P&L, it makes the desk accountable and its wins visible.

The financing waterfall

For every position, the desk asks one question: what is the cheapest legal way to hold this exposure, right now, given balance-sheet, margin, and counterparty constraints? The answer is a routing decision across venues.

> Financing waterfall — a ranked decision that routes each position to its cheapest financing venue — cash prime (margin loan), synthetic (total return swap / CFD), or repo — after accounting for the all-in spread, the margin it consumes, balance-sheet relief, and counterparty-concentration limits. The ranking is recomputed as spreads and limits move.

WORKED EXAMPLE

Routing $100m of equity exposure through the waterfall.
Benchmark SOFR = 5.00%. Compare three venues for one year of $100m long exposure:

Venue	Financing spread	Margin posted	All-in annual cost*
Cash prime (margin loan on $85m)	SOFR + 50bp	$15m equity	$5.425m
Total return swap (synthetic)	SOFR + 35bp	$10m IM	$5.850m
Repo-funded (if it were a bond)	SOFR + 20bp	2% haircut	cheapest where eligible

**All-in = financing interest + opportunity cost on posted margin at 5.00%.*
Here the cash route ($5.425m) beats the TRS ($5.850m) on pure cost, but the TRS frees balance sheet and ties up less margin ($10m vs $15m). The waterfall does not pick blindly on headline spread — it weighs the ~43bp cost gap against balance-sheet relief and counterparty limits. Shift the TRS spread 20bp tighter and the answer flips. Running this comparison continuously, across the whole book, is the daily work.

The best-in-class KPI dashboard

A world-class financing officer watches a small, disciplined set of numbers every morning. Each has a target and a trigger.

KPI	What it measures	Target / discipline
Weighted-avg financing spread	Blended spread over benchmark paid to finance the book	Minimise; beat prior quarter
Unencumbered cash	Cash free of any pledge, available same-day	≥ stressed one-day margin need
Days-to-liquidate	Time to exit the book without undue price impact	≤ redemption notice period
Margin-to-equity	Total IM/VM posted ÷ fund equity	Cap per mandate; monitor trend
Counterparty concentration	Net exposure to any single prime/dealer	Cap (e.g. ≤40% per prime)
Borrow cost vs benchmark	Short-book borrow fees vs a GC/benchmark rate	Minimise; renegotiate outliers

Real-time, what-if infrastructure

The dashboard is only as good as the data behind it. Best-in-class means what-if capability: before a PM puts on a trade, the desk can price its marginal impact on IM at each venue, on counterparty concentration, and on

the liquidity-coverage ratio — in seconds, not overnight. Margin is pre-computed with the desk's own SPAN and SIMM replicas; counterparty exposure is netted (not gross) and refreshed intraday; collateral and cash share one live inventory so the cheapest-eligible asset is always posted automatically.

> What-if margin — the ability to compute, before execution, how a prospective trade changes required IM at each clearing and bilateral venue, so the PM sees the true financing-adjusted return and the desk routes the trade to the cheapest-margin venue.

The maturity model — laggard to leader

Dimension	Laggard	Follower	Leader
Financing stance	Back-office cost	Cost monitored	Managed P&L / alpha
Margin	Takes CCP/FCM number	Reconciles daily	Pre-computes & what-ifs
Primes	Single prime	Two primes	Multi-prime, optimised routing
Cash	Idle at prime	Swept weekly	Swept nightly, netted exposure
Liquidity	Gut feel	Monthly stress	Daily coverage ratio ≥1.2×
Collateral	Posts cash always	Cheapest-eligible manually	Automated optimisation, one inventory

WORKED EXAMPLE

What "leader" is worth, in dollars, on a $5bn fund.

Lever	Improvement	Annual value
Financing-spread compression (multi-prime competition)	15bp on $3.5bn financed	$5.25m
Nightly cash sweep into reverse repo	70bp on $400m idle	$2.80m
Collateral optimisation (cheapest-eligible)	Free up $200m HQLA at 3%	$6.00m
Borrow-fee recovery / renegotiation	Saves on short book	$1.50m
Total treasury-driven alpha		$15.55m / year

On a $5bn fund, becoming a leader rather than a laggard is worth roughly $15.5m a year — about 31bp of fund return — delivered with no additional market risk. That is why the best shops treat the financing desk as a profit centre and staff it accordingly.

Governance

A leader pairs optimisation with robust governance: a counterparty-risk committee that sets and enforces per-prime caps; a liquidity-risk policy with a daily coverage ratio and a documented contingency funding plan; rehypothecation limits and segregation terms negotiated into every prime brokerage agreement; and clear escalation when any KPI breaches its trigger. Optimisation without governance is how LTCM, Archegos, and LDI happened.

Becoming a recognised expert

The individual path to mastery is concrete. The vocabulary: fluency in IM/VM, SPAN/SIMM, UMR/AANA, rehypothecation and segregation, the financing waterfall, cheapest-to-deliver, and the liquidity-coverage ratio — used precisely, not loosely. The frameworks: the cause→mechanism→lesson lens on every crisis; the "who lends, who borrows, where does collateral sit, what happens when everyone wants cash at once" questions; the all-in financing cost comparison on every trade. The canon: read the primary sources — ISDA documentation and SIMM methodology, CCP margin rulebooks, Basel liquidity standards (LCR/NSFR), Federal Reserve repo and reserve data, and the central-bank financial-stability reports that flag the next stress early.

> Treasury is not the department that pays for the trade. It is the department that decides whether the trade is worth having after the cost of financing it — and whether the fund survives the day the trade goes wrong.
>
> — the financing-desk credo

BEST IN CLASS

The capstone in one breath: run financing as a P&L with a daily dashboard (weighted-average spread, unencumbered cash, days-to-liquidate, margin-to-equity, counterparty concentration, borrow-cost vs benchmark); route every position through a financing waterfall to its cheapest venue; pre-compute and what-if margin and counterparty exposure in real time off one live cash-and-collateral inventory; sweep all idle cash nightly and post only cheapest-eligible collateral; run a daily liquidity-coverage ratio of ≥1.2× against a worst-case stress; govern it with per-prime caps, rehypothecation limits, and a contingency funding plan; and measure the whole thing — because a leader desk is worth tens of basis points of pure, repeatable alpha, and a laggard desk is one bad day from the war-stories chapter.

13 · REFERENCE

Glossary & Further Reading

A fast reference to the vocabulary used across this guide, followed by the canon a serious financing professional is expected to know.

A. Glossary of key terms

Prime brokerage & financing

Prime broker (PB) — the bank that provides a fund with custody, clearing, financing, securities lending, reporting and capital introduction under one relationship.

Margin loan — the cash a PB lends against long positions, charged at a spread over SOFR; the main way a fund finances cash longs.

Financing spread — the premium over the benchmark (e.g. SOFR + 40bp) a fund pays to borrow, or earns as a rebate when short.

Short rebate — the interest a fund earns on the cash collateral backing a stock borrow; high for easy-to-borrow names, negative for “specials”.

Stock borrow fee / specials — the fee to borrow a security to sell short; a “special” is a hard-to-borrow name commanding a large fee.

Gross vs net exposure — gross is longs plus shorts (the total financed footprint); net is longs minus shorts (directional risk). Gross drives financing cost.

Capital introduction (“cap intro”) — a PB service connecting the fund to prospective investors.

Margin & collateral

Regulation T — the US rule setting 50% initial and 25% maintenance margin on equities in a standard account.

Portfolio margin (TIMS) — a risk-based method that nets longs, shorts and hedges and sizes margin to the portfolio's stress loss, usually far below Reg-T.

House margin / add-on — margin a PB charges above the regulatory minimum, often for concentration, liquidity or stress.

SPAN — the Standard Portfolio Analysis of Risk, the scenario-based margin model for futures and listed options.

Haircut / initial margin (repo) — the over-collateralisation a repo lender requires; a 2% haircut funds 98% of a bond's value.

Variation margin (VM) / Initial margin (IM) — VM covers daily mark-to-market; IM covers potential future exposure during a close-out.

SIMM — the ISDA Standard Initial Margin Model used to compute regulatory IM on uncleared derivatives.

UMR / AANA — Uncleared Margin Rules; the Average Aggregate Notional Amount test that determines which funds must post two-way IM.

Rehypothecation — a PB's reuse of assets a fund posts; the practice that trapped client assets when Lehman failed.

Segregated IM — regulatory initial margin held at a third-party custodian, bankruptcy-remote and not reusable by the counterparty.

Instruments, repo & risk

Repo / reverse repo — borrowing cash against securities (repo) or lending cash against securities (reverse); the fund's bond-financing workhorse.

General collateral (GC) vs special — GC trades at the market repo rate; a special trades rich because a specific bond is in demand.

Sponsored repo / FICC — central-cleared repo that lets funds trade with a wider set of counterparties through a sponsoring member.

Basis trade — a leveraged trade capturing the small spread between a cash Treasury and its future, funded in repo at very high leverage.

Total return swap (TRS) — a swap in which the fund receives the total return of an asset and pays a financing leg (e.g. SOFR + spread); synthetic leverage without owning the asset.

Delta-one — the desk that provides synthetic exposure (swaps, futures, ETFs) that moves one-for-one with the underlying.

FCM (Futures Commission Merchant) — the clearing broker that stands between a fund and a derivatives clearing house (CCP).

CCP — a Central Counterparty that novates cleared trades and collects initial and variation margin.

Multi-prime — running several prime brokers to diversify counterparty risk, win financing competition and expand borrow access.

Days-to-liquidate — how long it would take to exit a position without undue market impact; the core asset-liquidity measure.

Gate / lock-up / side pocket — investor-side tools that limit or delay redemptions, shaping the fund's funding stability.

Margin-to-equity — total margin posted divided by fund equity; a headline measure of leverage intensity.

B. Further reading — the canon

- Marcia Stigum & Anthony Crescenzi, *Stigum's Money Market* — the definitive reference on repo and securities financing.
- Jon Gregory, *The xVA Challenge* — counterparty risk, margin and collateral from the practitioner's seat.
- Zoltan Pozsar — writing on the plumbing of repo, collateral and leverage that moves these markets.
- Darrell Duffie — research on dealer balance sheets, repo runs and the economics of intermediation.
- Credit Suisse Group Special Committee, *Report on Archegos Capital Management* (2021) — the definitive case study in concentrated synthetic leverage and margin failure.
- Roger Lowenstein, *When Genius Failed* — the LTCM story and the anatomy of a leverage-driven liquidity spiral.
- BIS & Federal Reserve notes on the March 2020 Treasury-basis unwind and on relative-value leverage — essential for understanding funding-driven dislocations.

DISCLAIMER

This guide is an educational reference. Figures in worked examples are illustrative and chosen to teach method, not to quote any specific fund or broker. Margin parameters (Reg-T, portfolio-margin stresses, SPAN scan ranges, SIMM weights, haircuts) are simplified for teaching and vary by broker, product and regime; always confirm current terms before relying on them.